Adaptive Retransmission Scheduling in Underwater Acoustic Sensor Networks based on ARQ_HMAC

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Abstract

Underwater Acoustic Sensor Networks (UASNs) are employed for many applications such as ocean monitoring, environmental observation, underwater exploration, and surveillance. Unlike terrestrial wireless networks, underwater communication relies mainly on acoustic signals. As acoustic signals have very low propagation speed and narrow bandwidth, the communication delay is large [2, 3, 4, 5]. The design of an efficient and reliable communication system for underwater networks is therefore difficult. One of the major issues in UASNs is packet collision. When multiple nodes send their packets while competing for the channel simultaneously, the packets may collide and fail to reach the destination. In such cases, the packets need to be retransmitted. However, multiple retransmissions cause increased delay, channel contention, and energy wastage in the network [3, 5, 9]. The large propagation delay of acoustic signals causes an additional difficulty, as retransmission schedules that perform well in terrestrial networks may not be suitable for underwater networks [2, 8]. This project is based on the ARQ_HMAC protocol, which is specifically designed for underwater acoustic sensor networks [1]. It uses contention-based channel access together with Automatic Repeat reQuest (ARQ) and a TDMA-based retransmission mechanism [1]. During normal communication, nodes

contend for the channel. If a transmission attempt fails, instead of immediately resuming contention for the channel, the node estimates the propagation delay and schedules the retransmission in a time slot different from that of the first transmission according to the estimated propagation delay. This helps reduce the probability of repeated retransmissions and collisions [1, 6, 8]. One drawback of ARQ_HMAC is that the propagation delay is estimated only once during the initial network setup [1]. However, the underwater communication conditions may change over time. Thus, the initial estimate of the propagation delay may no longer remain valid. Several studies on underwater communication have shown that underwater channel and propagation conditions can vary dynamically. Hence, adaptive techniques are required to cope with changing channel conditions [7, 3]. This may otherwise result in inefficient retransmission timing, more collisions, increased delay, and additional retransmissions. The goal of this project is to modify the retransmission process so that it can adapt to changing channel conditions in underwater communication. Instead of using only the initial estimated value of the round-trip time (RTT), the proposed scheme will continuously update the estimate based on the observed communication behavior and trigger retransmission using the updated value when an RTT change is detected. The original ARQ_HMAC protocol and the proposed adaptive protocol will be implemented in the UnetStack simulation framework. The relative performance of the two protocols will be evaluated in terms of Packet Delivery Ratio (PDR), throughput, number of retransmissions, collision rate, and end-to-end delay [1, 7, 6].

References

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